

# Open-World Anomaly Discovery for Adaptive AML

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## 1. Problem Statement

Traditional Anti-Money Laundering (AML) models rely heavily on **closed-world assumptions**. They are trained on historical data to detect known fraud typologies (e.g., layering, smurfing, cycle transfers). However, financial crime is highly adversarial and constantly evolving. When criminals introduce completely novel laundering strategies (e.g., new decentralized finance schemes or unseen AI-assisted shell company networks), traditional supervised models either fail to detect them entirely or confidently misclassify them into incorrect known categories.

The goal of this research is to build an intelligent, adaptive AML framework that can accurately flag known laundering behaviors while simultaneously discovering **unseen, emerging money laundering typologies** dynamically, and learning them incrementally without forgetting past knowledge.

## 2. What Are We Doing?

We are designing a **Continual Graph Learning** system for Open-World Anomaly Discovery. Instead of treating transactions as isolated events in a tabular dataset, we model the financial ecosystem as an interconnected graph. We then use a Graph Neural Network (GNN) combined with Open-Set Recognition to identify anomalous subgraphs that deviate from known patterns. These "unknowns" are clustered into new typologies and incrementally fed back into the model via Continual Learning. Finally, an Explainable AI (XAI) module translates these complex graph anomalies into readable reports for compliance officers.

## 3. The Dataset

We are utilizing the **PaySim Dataset** ( `PS_20174392719_1491204439457_log.csv` ).

- **Scale:** Over 6.3 million transactions.
- **Temporal Aspect:** Contains a `step` column representing time (1 step = 1 hour), spanning 30 days (743 steps).
- **Features:** Transaction type ( `PAYMENT` , `TRANSFER` , `CASH_OUT` , etc.), amount, source account, destination account, and balances before/after the transaction.

- **Ground Truth:** Includes an `isFraud` column identifying malicious behavior, which is highly imbalanced (only a small fraction are fraudulent).

## 4. Preprocessing & Feature Engineering

Before feeding data into a graph, we must engineer features to give the GNN rich context:

- **Node (Account) Features:**
  - Transaction frequency (velocity).
  - In-degree and out-degree (number of unique senders/receivers).
  - Average transaction size and balance delta over time windows.
- **Edge (Transaction) Features:**
  - Normalized transaction amount.
  - Transaction type (one-hot encoded).
  - Time of day/week (derived from the `step` feature).
- **Graph Construction:**
  - **Nodes:** Entities/Accounts (`nameOrig`, `nameDest`).
  - **Edges:** Transactions connecting accounts.
  - *Optimization:* Constructing **Temporal/Dynamic Graph Snapshots** (e.g., daily or hourly graphs) to capture how laundering networks form and dissipate over time.

## 5. The Optimized Pipeline Flow

Your initial flow was excellent, but I have slightly optimized it to ensure features are prepared before graph construction, and to represent the branching nature of Open-Set recognition:

```
graph TD
    A[Raw Transaction Data] --> B[Feature Engineering Node/Edge]
    B --> C[Temporal Graph Construction]
    C --> D[Temporal GNN Encoder]
    D --> E{Open-Set Evaluator <br/> Energy/Distance}
    E -- Matches Known Distribution --> F[Known AML Classification]
    E -- Deviates from Distribution --> G[Unknown Anomaly Pool]
    G --> H[Density-Based Clustering <br/> HDBSCAN]
    H --> I[Human-in-the-Loop Verification]
    I --> J[Continual Learning Update]

    F --> K[Explainable AML Alert]
    H --> K
```

## 6. Recommended Models & Approaches

- **GNN Encoder: TGN (Temporal Graph Networks) or GraphSAGE.** Since AML relies heavily on the *timing* of fast money movement, a temporal GNN is far superior to a static one.
- **Open-Set Detection: Energy-Based Models (EBM) or Deep SVDD.** Instead of a standard Softmax layer, energy scores are better at identifying inputs that do not belong to any known training distribution.
- **Clustering: HDBSCAN.** Perfect for this use case because it doesn't force every anomaly into a cluster (handles noise well) and works natively with varying density in high-dimensional embedding spaces.
- **Continual Learning: Experience Replay** (keeping a memory bank of past typologies) paired with **EWC (Elastic Weight Consolidation)** to penalize changes to weights that are important for old typologies, preventing catastrophic forgetting.
- **Explainable AI (XAI): GNNExplainer or PGM-Explainer.** These will output a specific subgraph (e.g., highlighting a specific 4-node ring of shell companies) to show the analyst *why* the transaction is flagged.

## 7. Expected Results

1. **High Accuracy on Known Typologies:** Maintaining state-of-the-art AUROC, AUPRC, and F1 scores on the typologies the model was originally trained on.
2. **Robust Discovery (Open Set F1):** High Unknown Detection Rate (successfully flagging new laundering patterns without misclassifying them as benign).
3. **Meaningful Clusters (ARI / NMI):** The clustering algorithm groups mathematically similar anomalies into distinct, actionable typologies for analysts.
4. **Zero Forgetting:** The Continual Learning module shows a near-zero "Forgetting Rate" across multiple update cycles.
5. **Actionable Intelligence:** Graph-visualized alerts that significantly reduce analyst investigation time.